



E-Commerce Sales Analysis

SQL • Data Cleaning • Exploratory Data Analysis (EDA) • Data Visualization
• Business Insight

Project Context :

E-commerce transaction data can provide valuable information about sales performance, products, customers, and profitability. Analyzing transaction data helps identify patterns and generate insights that provide a better understanding of business performance.

Primary Goal :

Analyze e-commerce transaction data using SQL to evaluate sales performance, product and category contributions, customer spending, regional performance, and profitability.

Data Foundation:

5,000 Transactions

14 Variables

PostgreSQL + SQL

Excel → CSV → PostgreSQL

Key Deliverables :

Sales Performance Analysis

Product & Category Analysis

Customer Analysis

Regional Analysis

Payment Method Analysis

Profit Analysis

Time-based Sales Analysis

Database Structure

Database Setup

Database: ecommerce_analysis

DBMS: PostgreSQL

Schema: public

Table: sales

Database Workflow

Excel → CSV → PostgreSQL →
SQL Analysis

Table Structure

The sales table contains 5,000 e-commerce transaction records with 14 columns covering transaction, customer, product, sales, and payment information.

Column	Description
order_id	Unique order identifier
order_date	Date of the transaction
customer_name	Customer name
region	Customer region
city	Customer city
category	Product category
sub_category	Product sub category
Product name	Product name
quantity	Number of items purchased
unit_price	Price per item
discount	Discount applied
sales	Total sales value
profit	Profit generated
payment_mode	Payment method

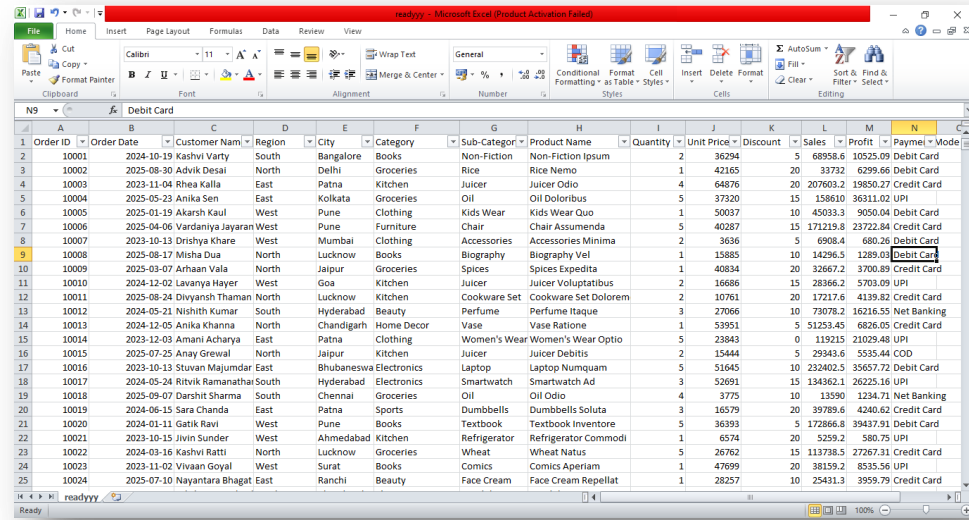
Database Preparation & Cleaning

Data Preparation

- Reviewed the dataset structure and column names.
- Verified the 14 columns required for analysis.
- Checked date and numerical data formats.
- Confirmed the dataset contained 5,000 transaction records.
- Prepared the dataset as a CSV file for PostgreSQL import.

Data Quality Check

- Checked for missing values. Verified the consistency of categorical values.
- Checked numerical fields such as Quantity, Unit Price, Discount, Sales, and Profit.
- Verified the imported data in PostgreSQL.



Order ID	Order Date	Customer Name	Region	City	Category	Sub-Category	Product Name	Quantity	Unit Price	Discount	Sales	Profit	Payment Mode
10001	2024-10-19	Kashvi Varty	South	Bangalore	Books	Non-Fiction	Non-Fiction Ipsum	2	36294	5	68958.6	10525.09	Debit Card
10002	2025-08-30	Advik Desai	North	Delhi	Groceries	Rice	Rice Nemo	1	42165	20	33732	6299.66	Debit Card
10003	2023-11-04	Rhina Kalla	East	Patna	Kitchen	Juicer	Juicer Odio	4	64876	20	207603.2	19850.27	Credit Card
10004	2025-05-23	Anika Sen	East	Kolkata	Groceries	Oil	Oil Doloribus	5	37320	15	158610	36311.02	UPI
10005	2025-01-19	Akarsh Kaul	West	Pune	Clothing	Kids Wear	Kids Wear Quo	1	50037	10	45033.3	9050.04	Debit Card
10006	2025-04-06	Vardaniya Jayaran	West	Pune	Furniture	Chair	Chair Assumenda	5	40287	15	171219.8	23722.84	Credit Card
10007	2023-10-13	Drishya Khare	West	Mumbai	Clothing	Accessories	Accessories Minima	2	3636	5	6908.4	680.26	Debit Card
10008	2025-08-17	Misha Dua	North	Lucknow	Books	Biography	Biography Vel	1	15885	10	14296.5	1289.03	Debit Card
10009	2025-03-07	Arhaan Vala	North	Jaipur	Groceries	Spices	Spices Expedita	1	40834	20	32667.2	3700.89	Credit Card
10010	2024-12-02	Lavanya Hayer	West	Goa	Kitchen	Juicer	Juicer Voluptatibus	2	16686	15	28366.2	5703.09	UPI
10011	2025-08-24	Divyansh Thaman	North	Lucknow	Kitchen	Cookware Set	Cookware Set Dolorem	2	10761	20	12121.6	4139.82	Credit Card
10012	2024-05-21	Nishith Kumar	South	Hyderabad	Beauty	Perfume	Perfume Itaque	3	27066	10	73078.2	16216.55	Net Banking
10013	2024-12-05	Anika Khanna	Chandigarh	Home Decor	Vase	Vase Ratione	Vase Ratione	1	53951	5	51251.45	6826.05	Credit Card
10014	2023-12-03	Amani Acharya	East	Patna	Clothing	Women's Wear	Women's Wear Optio	5	23843	0	119215	12029.48	UPI
10015	2025-07-25	Anay Grewal	North	Jaipur	Kitchen	Juicer	Juicer Debitis	2	15444	5	29343.6	5535.44	COO
10016	2023-10-13	Stuvan Majumdar	East	Bhubaneswa	Electronics	Laptop	Laptop Numquam	5	51645	10	232402.5	35657.72	Debit Card
10017	2024-05-24	Ritvik Ramanathai	South	Hyderabad	Electronics	Smartwatch	Smartwatch Ad	3	52691	15	134362.1	26225.16	UPI
10018	2025-09-07	Darshit Sharma	South	Chennai	Groceries	Oil	Oil Odio	4	3775	10	13590	1234.71	Net Banking
10019	2024-06-15	Sara Chanda	East	Patna	Sports	Dumbbells	Dumbbells Soluta	3	16579	20	39789.6	4240.62	Credit Card
10020	2024-01-11	Gatik Ravi	West	Pune	Books	Textbook	Textbook Inventore	5	36393	5	172866.8	39437.91	Debit Card
10021	2023-10-15	Jivin Sunder	West	Ahmedabad	Kitchen	Refrigerator	Refrigerator Commodi	1	6574	20	5259.2	580.75	UPI
10022	2024-03-16	Kashvi Ratti	North	Lucknow	Groceries	Wheat	Wheat Natus	5	26762	15	113738.5	27267.31	Credit Card
10023	2023-11-02	Vivaan Goyal	West	Surat	Books	Comics	Comics Aperiam	1	47699	20	3819.2	8535.56	UPI
10024	2025-07-10	Nayantara Bhagat	East	Ranchi	Beauty	Face Cream	Face Cream Repellat	1	28257	10	25431.3	3959.79	Credit Card

Data Validation

5,000 records successfully imported into PostgreSQL.

Excel → Data Cleaning → CSV → PostgreSQL → 5,000 Records

KEY BUSINESS METRICS

533,666,024.35

Total Sales

5,000

TOTAL TRANSACTIONS

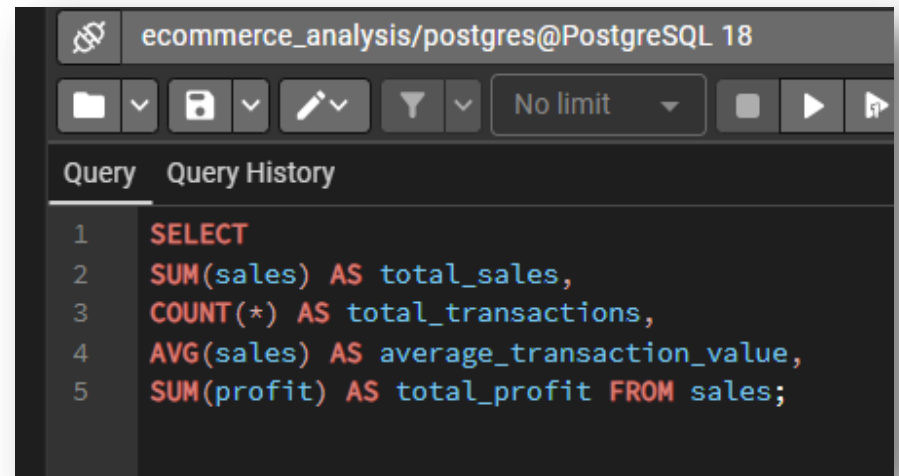
106,733.204

AVERAGE TRANSACTION VALUE

79,708,734.91

TOTAL PROFIT

SQL USED



The screenshot shows a PostgreSQL query editor interface. The title bar reads "ecommerce_analysis/postgres@PostgreSQL 18". Below the title bar is a toolbar with icons for file operations, a search filter, and a "No limit" dropdown. The main area is divided into "Query" and "Query History" tabs. The "Query" tab is active, displaying a SQL query with line numbers 1 through 5. The query is a SELECT statement that calculates total sales, total transactions, average transaction value, and total profit from a table named "sales".

```
1 SELECT
2 SUM(sales) AS total_sales,
3 COUNT(*) AS total_transactions,
4 AVG(sales) AS average_transaction_value,
5 SUM(profit) AS total_profit FROM sales;
```

Sales by Category

Q. 1. What is the total revenue?

The total revenue generated from 5,000 transactions is Rp533,666,024.35.

Query	Query History
1	SELECT
2	SUM (sales) AS total_revenue
3	FROM sales;
4	

Data Output	Messages	Explain	×	Graph Visu
         				
	total_revenue numeric			
1	533666024.35			

Q. 2. How many transactions are recorded?

A total of 5,000 transactions are recorded in the dataset.

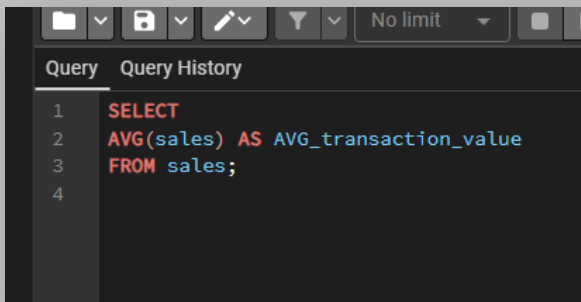
Query	Query History
1	SELECT
2	COUNT(*) AS total_transaction
3	FROM sales;
4	

Data Output	Messages	Explain	×	Graph Visualiser	×
         					
	total_transaction bigint				
1	5000				

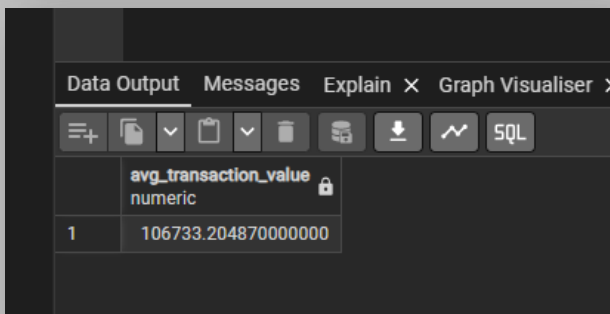
Sales by Category

Q. 3. What is the average transaction value?

Average Transaction Value: Rp106,733.20
Each transaction generates an average sales value of approximately Rp106,733.20.



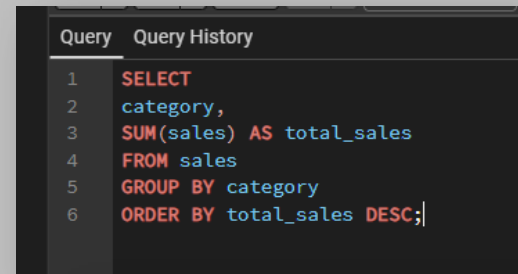
```
1 SELECT
2 AVG(sales) AS avg_transaction_value
3 FROM sales;
4
```



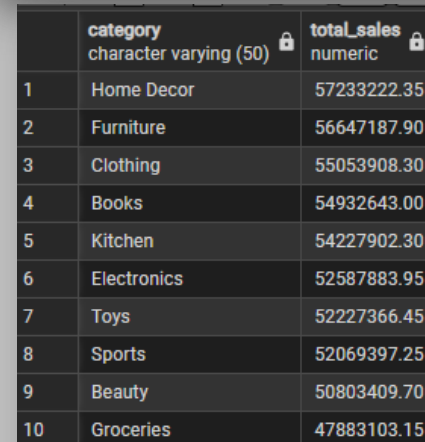
	avg_transaction_value numeric
1	106733.204870000000

Q. 4. Which product category generates the highest total sales?

Home Decor leads all product categories with total sales of Rp57,233,222.35.



```
1 SELECT
2 category,
3 SUM(sales) AS total_sales
4 FROM sales
5 GROUP BY category
6 ORDER BY total_sales DESC;
```



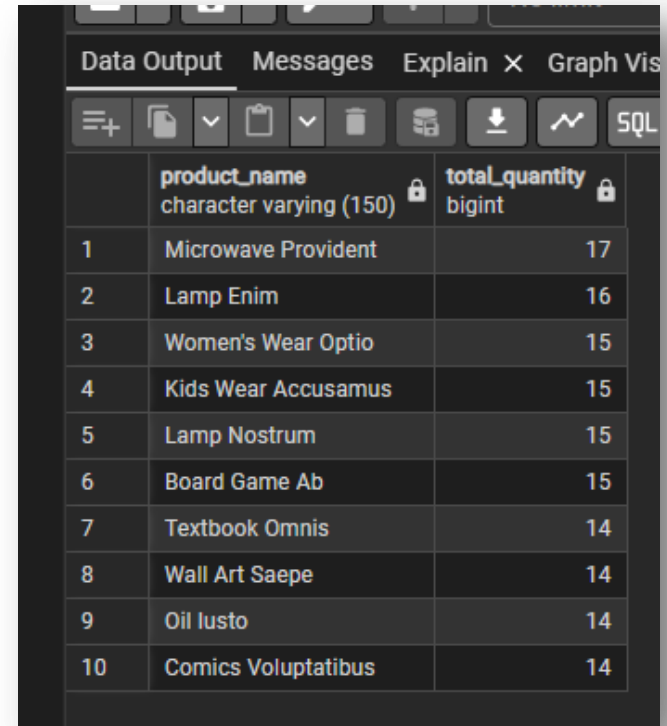
	category character varying (50)	total_sales numeric
1	Home Decor	57233222.35
2	Furniture	56647187.90
3	Clothing	55053908.30
4	Books	54932643.00
5	Kitchen	54227902.30
6	Electronics	52587883.95
7	Toys	52227366.45
8	Sports	52069397.25
9	Beauty	50803409.70
10	Groceries	47883103.15

Sales by Category

Q. 5. Which products have the highest total sales?

Based on the analysis, Microwave has the highest total sales among all products, amounting to 17. This indicates that Microwave makes the largest contribution to product sales in the dataset.

```
Query  Query History
1  SELECT
2  product_name,
3  SUM(quantity) AS total_quantity
4  FROM sales
5  GROUP BY product_name
6  ORDER BY total_quantity DESC
7  LIMIT 10;
```



	product_name character varying (150)	total_quantity bigint
1	Microwave Provident	17
2	Lamp Enim	16
3	Women's Wear Optio	15
4	Kids Wear Accusamus	15
5	Lamp Nostrum	15
6	Board Game Ab	15
7	Textbook Omnis	14
8	Wall Art Saepe	14
9	Oil Iusto	14
10	Comics Voluptatibus	14

Sales by Category

Q. 6. Which product generates the highest revenue?

Based on the analysis, Headphones Accusanti generates the highest revenue, amounting to 857,184.20. This indicates that the product contributes the largest amount of revenue compared to other products in the dataset.

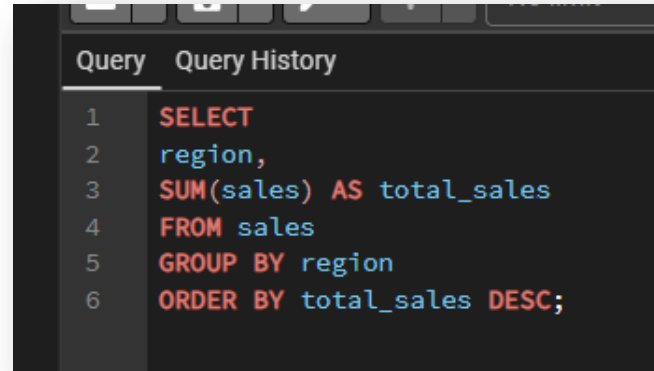
```
Query  Query History
1  SELECT
2  product_name,
3  SUM(sales) AS total_sales
4  FROM sales
5  GROUP BY product_name
6  ORDER BY total_sales DESC
7  LIMIT 10;
8
```

	product_name character varying (150)	total_sales numeric
1	Headphones Accusanti...	857184.20
2	Spices Quibusdam	687651.25
3	Accessories Repellendus	687083.30
4	Bed Tenetur	670994.80
5	Laptop Similique	666640.60
6	Lamp Enim	665920.20
7	Shampoo Commodi	659546.60
8	Lamp Veritatis	652909.50
9	Men's Wear Voluptatibus	649048.00
10	Textbook Omnis	633246.25

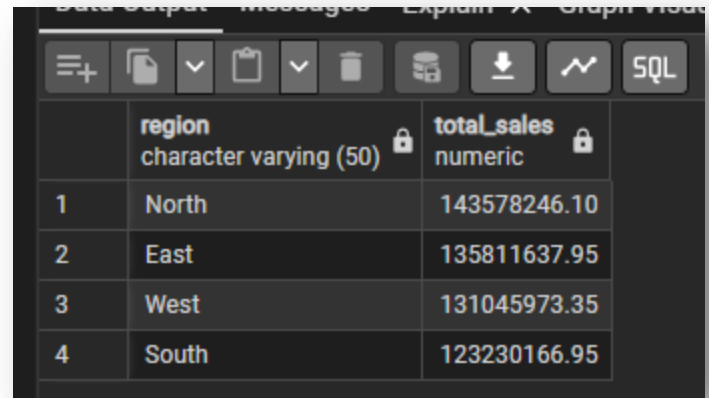
Sales by Category

Q. 7. Which region generates the highest total sales?

The North region records the highest total sales at Rp143,578,246.10, followed by East at Rp135,811,637.95, West at Rp131,045,973.35, and South at Rp123,230,166.95. North's sales are approximately Rp20.35 million higher than South, which has the lowest total sales. This indicates differences in sales contributions across regions, with North being the largest contributor in the dataset. However, further analysis of transaction volume and average transaction value is needed to understand what drives this difference.



```
Query  Query History
1  SELECT
2  region,
3  SUM(sales) AS total_sales
4  FROM sales
5  GROUP BY region
6  ORDER BY total_sales DESC;
```



	region character varying (50)	total_sales numeric
1	North	143578246.10
2	East	135811637.95
3	West	131045973.35
4	South	123230166.95

Sales by Category

Q. 8. Who are the top 10 customers by spending?

Aaryahi Madan is the highest-spending customer, with total spending of Rp650,151.90. The top 10 customers' spending ranges from Rp426,563.60 to Rp650,151.90, showing variation in spending among the highest-spending customers. This analysis helps identify customers with the largest spending contributions and can support customer retention efforts.

```
Query  Query History
1  SELECT
2  customer_name,
3  SUM(sales) AS total_spending
4  FROM sales
5  GROUP BY customer_name
6  ORDER BY total_spending DESC
7  LIMIT 10;
```

Data Output Messages Explain X Graph Visuali		
	customer_name character varying (100)	total_spending numeric
1	Aaryahi Madan	650151.90
2	Kimaya Vala	490935.00
3	Badal Rao	485698.00
4	Adira Dhingra	479389.00
5	Sumer Mann	469937.00
6	Nayantara Majumdar	459763.50
7	Prerak Shere	451862.40
8	Bhamini Sane	441410.60
9	Samarth Wagle	427715.50
10	Arhaan Basu	426563.60

Sales by Category

Q. 9. Which payment methods are used most frequently?

Net Banking is the most frequently used payment method, with 1,010 transactions, followed by COD with 1,005 transactions and Debit Card with 1,003 transactions. Meanwhile, UPI has the lowest number of transactions, at 988. The difference between the most and least frequently used methods is only 22 transactions, indicating that the use of the five payment methods is relatively evenly distributed in the dataset.

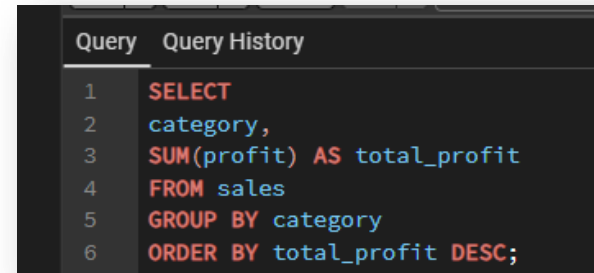
```
Query  Query History
1  SELECT
2  payment_mode,
3  COUNT(*) AS total_transactions
4  FROM sales
5  GROUP BY payment_mode
6  ORDER BY total_transactions DESC;
```

	payment_mode character varying (50)	total_transactions bigint
1	Net Banking	1010
2	COD	1005
3	Debit Card	1003
4	Credit Card	994
5	UPI	988

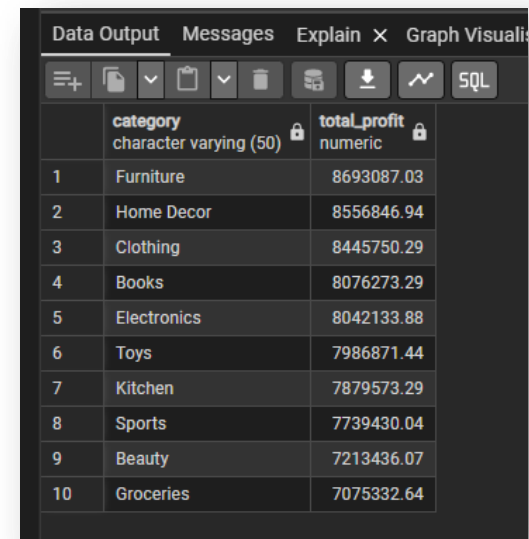
Sales by Category

Q. 10. What is the total profit by category?

Based on the analysis, Furniture generates the highest total profit at Rp8,693,087.03, followed by Home Decor at Rp8,556,846.94 and Clothing at Rp8,445,750.29. Meanwhile, Groceries records the lowest total profit at Rp7,075,332.64. The profit difference between the highest- and lowest-performing categories is Rp1,617,754.39. This indicates that Furniture contributes the largest total profit in the dataset, although profit differences should be considered alongside sales and cost data to better understand the factors influencing profitability.



```
Query  Query History
1  SELECT
2  category,
3  SUM(profit) AS total_profit
4  FROM sales
5  GROUP BY category
6  ORDER BY total_profit DESC;
```



	category character varying (50)	total_profit numeric
1	Furniture	8693087.03
2	Home Decor	8556846.94
3	Clothing	8445750.29
4	Books	8076273.29
5	Electronics	8042133.88
6	Toys	7986871.44
7	Kitchen	7879573.29
8	Sports	7739430.04
9	Beauty	7213436.07
10	Groceries	7075332.64

Key Findings & Recommendations

Key Findings

- Home Decor recorded the highest total sales at Rp57,233,222.35.
- North recorded the highest total sales among all regions, at Rp143,578,246.10.
- Aaryahi Madan was the highest-spending customer, with total spending of Rp650,151.90.
- Net Banking was the most frequently used payment method, with 1,010 transactions.
- Furniture generated the highest total profit, at Rp8,693,087.03.

Recommendations

- Evaluate the performance of the Home Decor category to understand the factors contributing to its high sales.
- Review sales distribution across regions to identify opportunities for improvement in regions with lower sales.
- Consider customer retention programs based on customer spending patterns.
- Maintain a variety of payment methods, as the five methods were used at relatively similar frequencies.

Conclusion & Limitations

Conclusion

- The analysis was conducted on 5,000 transactions using PostgreSQL and SQL.
- The analysis covered sales performance, products, customers, regions, payment methods, and profitability.
- The results provide an overview of sales and profit distribution within the dataset.

Limitations

- The analysis was based solely on the variables available in the dataset.
- The dataset does not provide sufficient information to explain the factors behind differences in sales and profit.
- The findings reflect this specific dataset and may not represent the entire e-commerce industry.